

External Validation of Structural Modes

MAAT Defect Features on Public scikit-learn Datasets

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Abstract

Papers 52–53 tested cross-domain structural transfer using repository-internal benchmarks. While useful, those experiments remained partly circular. This paper presents the first deliberately external validation of MAAT structural modes on public `scikit-learn` datasets: breast cancer, wine recognition, and handwritten digits.

The predeclared target is sample-level hardness, defined as the fraction of misclassifications under repeated stratified cross-validation with a fixed Random Forest classifier. MAAT supports ($H, B, S, V, R_{\text{rob}}$) are computed solely from feature geometry and local label statistics. These are compared against strong external baselines including local label disagreement, kNN density, centroid distance, and PCA reconstruction error.

MAAT scalar cost and robustness loss show consistent positive correlations with sample hardness across all three datasets ($\rho_S \simeq 0.323\text{--}0.330$). However, they are outperformed by simple local label disagreement ($\rho_S \simeq 0.430$) and external geometric baselines ($\rho_S \simeq 0.389$). MAAT v1.4 field features beat shuffled-defect nulls on two of three datasets but do not dominate classical baselines. This first external benchmark therefore breaks the internal-artifact loop and reveals both measurable signal and clear limitations of the current structural mode implementation.

1 Purpose

The main limitation of the previous transfer papers was not their methodology, but their data source. Papers 52–53 used internal SAT, quantum, active-significance, and societal toy artifacts. Those benchmarks clarified the role of connectivity and mode separation, but they could not answer the stronger question:

Do MAAT structural supports carry information on datasets not generated by the MAAT framework?

The present paper is designed as a first external reality check. It deliberately uses small, public, well-understood datasets rather than new MAAT-generated toy targets. The goal is not to outperform all classical feature-engineering baselines. The goal is to test whether the support language retains nontrivial signal under an external sample-hardness target.

2 External Datasets

Three public datasets are loaded through `scikit-learn`. They differ in sample size, dimensionality, and number of classes.

Table 1: External datasets used in Paper 54.

Dataset	Samples	Features	Classes
Breast cancer	569	30	2
Wine	178	13	3
Digits	1797	64	10

These are not precision industrial benchmarks. They are intentionally small enough to make the protocol transparent and fully reproducible.

3 Predeclared Sample Hardness

For each dataset, features are standardised and a fixed Random Forest classifier is evaluated under repeated stratified five-fold cross-validation. For sample i , the primary hardness target is

$$h_i = \frac{1}{N_{\text{rep}}} \sum_{r=1}^{N_{\text{rep}}} \mathbf{1}[\hat{y}_i^{(r)} \neq y_i], \quad (1)$$

where $N_{\text{rep}} = 25$ in the reported benchmark. Each sample appears once per repetition in a held-out fold.

The mean cross-validation accuracies are:

Table 2: Hardness-target diagnostics.

Dataset	Mean accuracy	Mean h_i	Nonzero h_i fraction	Max h_i
Breast cancer	0.9586	0.0414	0.0756	1.0000
Wine	0.9802	0.0198	0.0449	1.0000
Digits	0.9383	0.0617	0.1252	1.0000

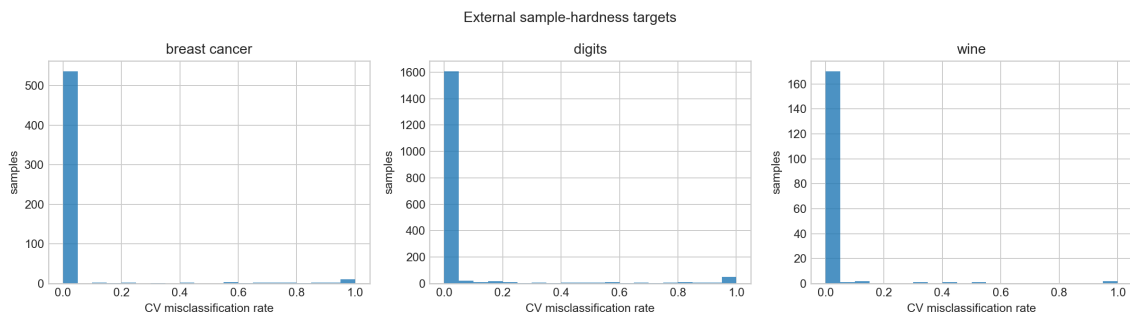


Figure 1: Distribution of repeated-CV sample-hardness values. The public datasets are comparatively clean, so the error-rate target is sparse, especially for wine.

4 MAAT Supports from Feature Geometry

The support construction is deliberately simple and predeclared. It should be read as a first heuristic operationalisation of the MAAT support language for tabular machine-learning data, not as a unique derivation from a microscopic theory of sample hardness. Let $N_k(i)$ denote the k nearest neighbours of sample i in standardised feature space. The local label distribution in

this neighbourhood defines a normalised entropy $E_i^{\text{label}} \in [0, 1]$. We set

$$H_i = 1 - E_i^{\text{label}}, \quad (2)$$

$$B_i = \max\left(0, p_i(y_i) - \max_{c \neq y_i} p_i(c)\right), \quad (3)$$

$$V_i = \text{norm}_{[0,1]}\left(\frac{1}{\bar{d}_k(i) + \epsilon}\right). \quad (4)$$

Here $p_i(c)$ is the neighbour-label fraction of class c , and $\bar{d}_k(i)$ is the mean kNN distance.

The activity support is not raw activity. It is a controlled local-spread window:

$$S_i = \exp\left[-\frac{1}{2} \left(\frac{\log(\bar{d}_k(i) + \epsilon) - m_{\log d}}{1.4826 \text{MAD}_{\log d} + \epsilon}\right)^2\right], \quad (5)$$

where $m_{\log d}$ and $\text{MAD}_{\log d}$ are the median and median absolute deviation of the log-spread distribution. This bounded definition is intentional: it avoids interpreting outlier spread or feature-space noise as automatically beneficial structure.

The robustness closure follows the v1.2.1 convention:

$$R_{\text{resp}}^{(i)} = (H_i B_i V_i)^{1/3}, \quad (6)$$

$$R_{\text{rob}}^{(i)} = \min\left(R_{\text{resp}}^{(i)}, (H_i B_i S_i V_i)^{1/4}\right). \quad (7)$$

The scalar MAAT cost is defined by

$$F_{\text{MAAT}}^{\text{scalar}}(i) = \frac{1}{5} \sum_{a \in \{H, B, S, V, R_{\text{rob}}\}} D_{a,i}, \quad D_{a,i} = -\log(\epsilon + \Gamma_{a,i}). \quad (8)$$

5 Baselines and Nulls

The benchmark includes several non-MAAT baselines:

- local label disagreement,
- kNN density anomaly,
- distance to class centroid,
- PCA reconstruction error,
- Isolation Forest anomaly score,
- simple feature norm and local spread.

For feature-set comparisons, a fixed Ridge regressor is evaluated by repeated five-fold cross-validation. No dataset-specific hyperparameter search is performed. The compared feature sets are:

- MAAT supports $(H, B, S, V, R_{\text{rob}})$,
- MAAT defects $(D_H, D_B, D_S, D_V, D_{R_{\text{rob}}})$,
- MAAT v1.4 field features with local defect, tail, and hotspot terms,
- external geometry baselines,
- label-geometry baselines,
- combined geometry baselines.

For the MAAT v1.4 field feature set, shuffled-defect nulls are generated by independently permuting each feature column before the same Ridge evaluation.

6 Results

6.1 Feature-Set Prediction

The primary result is the prediction of repeated-CV misclassification rate.

Table 3: Feature-set prediction of external sample hardness.

Feature set	Mean ρ_S	Median ρ_S	Mean AUROC	Mean R^2
External geometry	0.3889	0.4188	0.6685	0.1386
Combined geometry	0.3824	0.3933	0.6666	0.3595
MAAT supports	0.3241	0.3290	0.6393	0.2481
Label geometry	0.3107	0.3201	0.6438	0.3309
MAAT v1.4 field	0.2795	0.3393	0.6191	-0.4238
MAAT defects	0.2434	0.3672	0.5813	0.1331

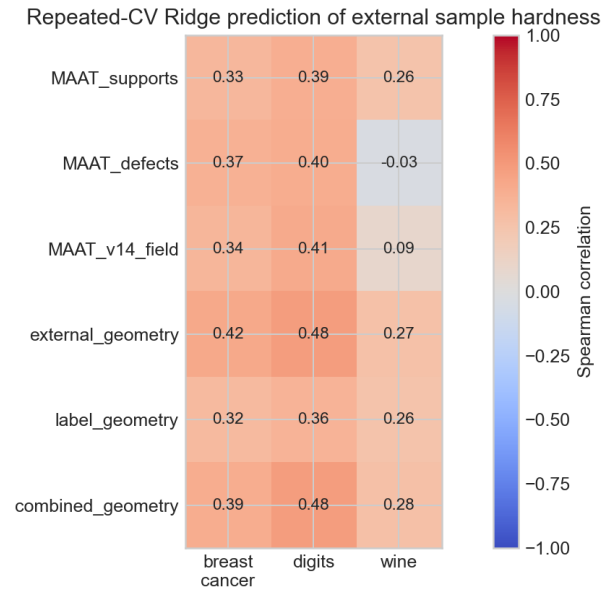


Figure 2: Dataset-level Spearman correlations for repeated-CV Ridge prediction of sample hardness.

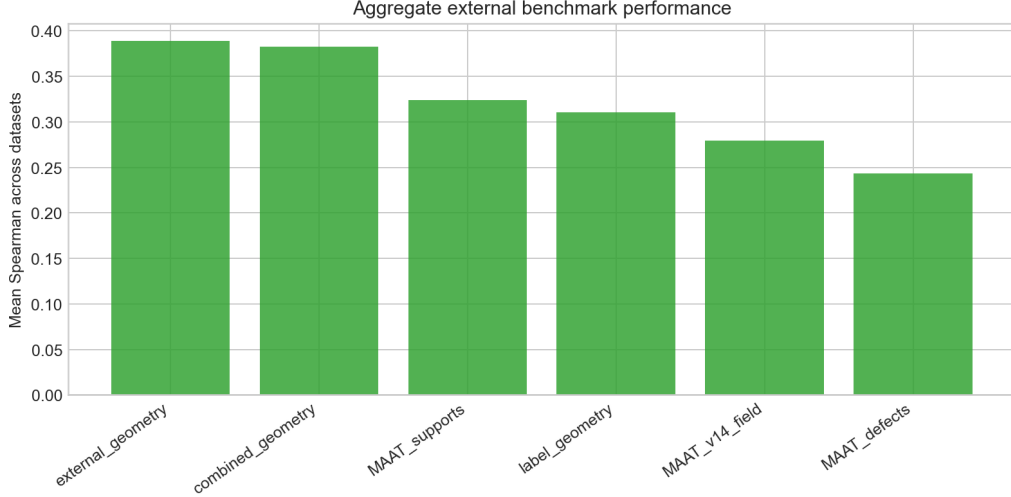


Figure 3: Mean cross-dataset Spearman correlation by feature set. Simple external geometry is strongest in this benchmark.

The external-geometry baseline is the best aggregate predictor. This is a useful negative result. It shows that the public-data sample-hardness task is not automatically solved by MAAT support language. The negative mean R^2 of the richer MAAT v1.4 field set suggests that this multi-scale representation may currently overparameterise small, clean public datasets.

6.2 Scalar Scores

The scalar-score comparison is also informative.

Table 4: Predeclared scalar scores versus external sample hardness.

Scalar score	Mean ρ_S	Median ρ_S	Mean AUROC
Label disagreement	0.4299	0.4539	0.6719
MAAT robustness loss	0.3301	0.3404	0.6708
MAAT scalar cost	0.3226	0.3197	0.6706
Centroid distance	0.2726	0.2308	0.6336
Density anomaly	0.1729	0.1721	0.5998
PCA reconstruction error	0.0521	0.0012	0.5439
Isolation anomaly	0.0361	0.0405	0.5364

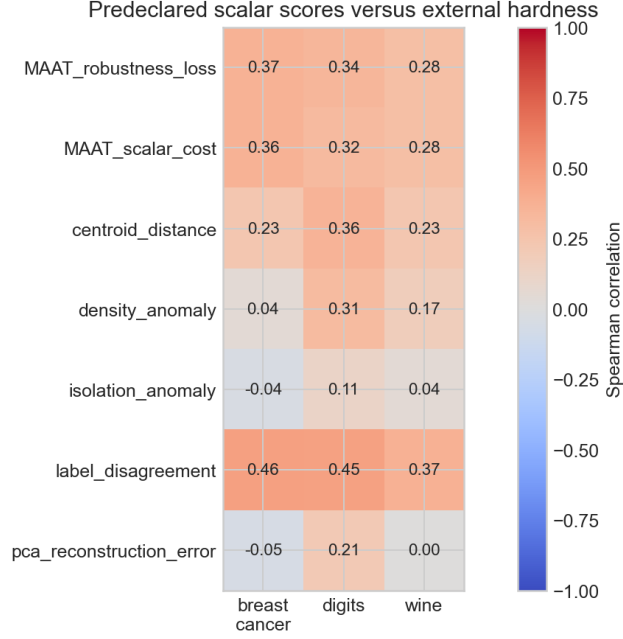


Figure 4: Predeclared scalar scores. MAAT scalar cost and robustness loss are consistently positive, but local label disagreement is stronger.

The scalar result is less negative for MAAT than the feature-set result. Both $F_{\text{MAAT}}^{\text{scalar}}$ and $1 - R_{\text{rob}}$ are positive across all three external datasets, with mean AUROC essentially tied with label disagreement. However, local label disagreement has the highest mean Spearman correlation.

6.3 Shuffled-Defect Nulls

The MAAT v1.4 field features are compared against shuffled-feature nulls.

Table 5: MAAT v1.4 field features against shuffled-defect nulls.

Dataset	Actual ρ_S	Null mean	Null p95	Margin vs p95
Breast cancer	0.3393	-0.0078	0.0724	0.2669
Wine	0.0925	0.0134	0.1703	-0.0778
Digits	0.4067	-0.0102	0.0326	0.3741

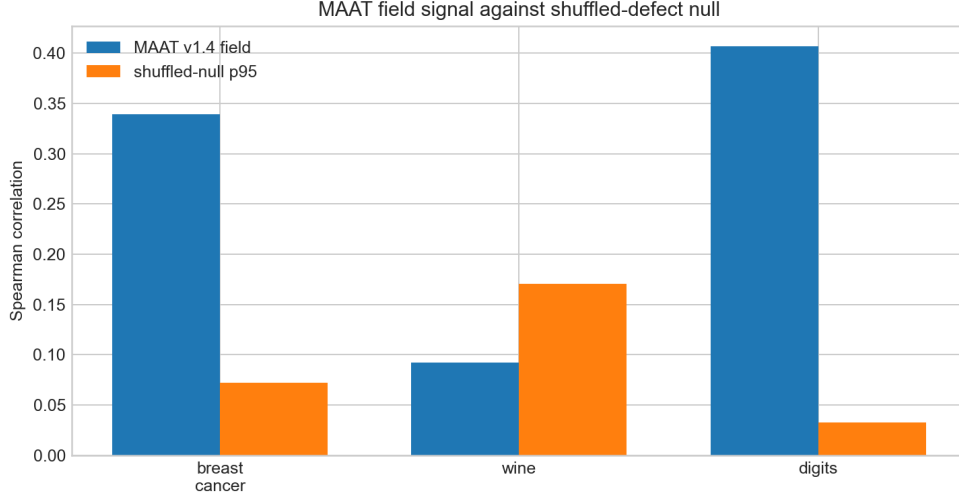


Figure 5: MAAT v1.4 field signal compared with shuffled-defect nulls. The field signal is above the 95th null percentile for breast cancer and digits, but not for wine.

This result is again mixed. The field representation carries non-random structure in two datasets, but the wine benchmark is too small and too clean for the present field features to clear the shuffled-null threshold.

7 Interpretation

Paper 54 produces a deliberately modest conclusion:

MAAT structural supports contain external sample-hardness signal, but they do not dominate simple geometric and label-local baselines in this first public-data benchmark.

The positive part is that scalar MAAT cost and robustness loss remain consistently aligned with sample hardness on external public data. The negative part is equally important: the richer v1.4 field representation does not yet outperform simple external geometry. Notably, the single strongest predictor in this benchmark is local label disagreement, a very simple nearest-neighbour statistic. This indicates that current MAAT supports still largely operate in the regime of classical local geometry and label information rather than capturing substantially novel structural modes.

8 Limitations

The limitations are important:

- The datasets are small, clean, and standard; they are not industrial ML benchmarks.
- Sample hardness is defined using Random Forest cross-validation, not an ensemble of modern model families.
- MAAT supports use labels through local neighbourhood statistics, so label-geometry baselines are essential and remain competitive.
- The v1.4 field feature map is predeclared but still phenomenological.
- No external hyperparameter search or large-scale benchmark suite is included.
- The result demonstrates limited external signal, not external universality.

9 Reproducibility

Repository folder:

`experiments/external_ml_hardness_paper54/`

Main script:

```
python3 experiments/external_ml_hardness_paper54/external_ml_hardness_paper54.py
```

The script writes:

- sample-level hardness and support tables,
- scalar-score summaries,
- feature-set performance tables,
- shuffled-null summaries,
- figures used in this paper,
- a JSON summary file.

Repository URL:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/external_ml_hardness_paper54

10 Data Attribution and License Note

The benchmark loads public datasets through `scikit-learn`. No raw external dataset files are re-distributed in the experiment folder. The CSV, JSON, and PNG files generated by the script are derived reproducibility artifacts. When reusing or discussing the benchmark, cite `scikit-learn` and the original dataset sources as appropriate. No endorsement by the `scikit-learn` maintainers, UCI dataset curators, or original dataset authors is implied.

11 Conclusion

This paper is the first MAAT structural-mode benchmark on fully external public machine-learning datasets. It gives neither a failure nor a triumph. Instead, it gives a useful boundary: MAAT scalar supports and robustness loss do carry external sample-hardness signal, but simple geometric and label-local baselines remain stronger in the current benchmark.

The next step should use harder external benchmarks, such as SAT Competition CDCL conflict counts, TSPLIB/TSP instances, graph-property prediction, or larger OpenML classification tasks. Future benchmarks should increase structural complexity rather than only sample size, because multi-scale defect-field features are expected to become more relevant in domains with genuine local frustration, bottlenecks, or combinatorial structure.

References

- [1] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research* **12**, 2825–2830 (2011).
- [2] L. Breiman, “Random Forests,” *Machine Learning* **45**, 5–32 (2001).

- [3] D. Dua and C. Graff, *UCI Machine Learning Repository*, University of California, Irvine, School of Information and Computer Sciences (2019).
- [4] C. Krieg, “Cross-Domain Transfer of Structural Selection: A Frozen-Weight Two-Domain Pilot Study,” MAAT Structural Selection Series, Paper 52, preprint (2026).
- [5] C. Krieg, “Robustness of Structural Mode Transfer in MAAT v1.4: Alternative Frozen Fits, Regularisation Tests, and Multi-Domain Null Validation,” MAAT Structural Selection Series, Paper 53, preprint (2026).